

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. Some nodes are highlighted with blue circles, and others with blue dots. The lines are thin and gray, creating a mesh-like structure.

Computer Networking

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a network of nodes and connections, with several nodes highlighted by blue circles and blue dots. The overall style is clean and modern, with a light gray background.

Chapter 1: Computer Networks and the Internet

Overview/roadmap:

- ◎ What *is* the Internet?
- ◎ What *is* a protocol?
- ◎ **The Network edge:** hosts, access network, physical media
- ◎ **The Network core:** packet/circuit switching, internet structure
- ◎ **Performance:** loss, delay, throughput
- ◎ **Security**
- ◎ **Protocol layers, service models**

Network and Computer Networks

A decorative network diagram in the top right corner of the slide. It features a complex web of interconnected nodes and lines. The nodes are represented by circles of varying sizes, some solid and some hollow, connected by thin, light gray lines. The overall structure is a dense, branching network.

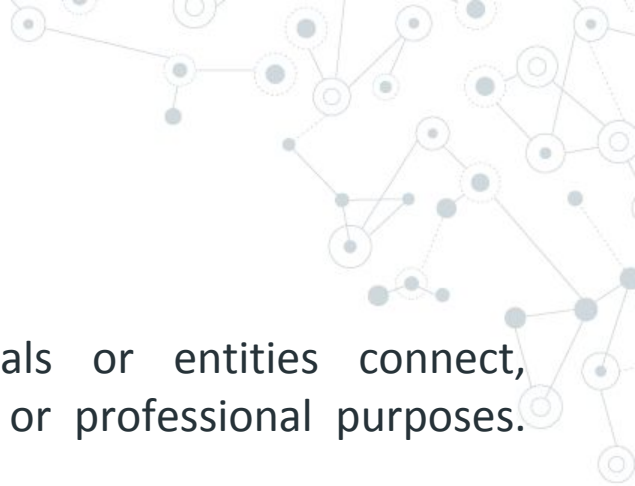
Network:

A collection of interconnected elements or nodes that communicate and interact with each other to achieve a specific purpose or to facilitate the exchange of information, resources, or services. Networks can take various forms and serve different functions.

Computer Network:

A system of interconnected computers and devices that enable data and information sharing, such as the internet or a local area network (LAN).

Different Type of Networks

A decorative network diagram in the top right corner of the slide. It consists of numerous small circles (nodes) of varying shades of gray and blue, connected by thin, light gray lines (edges). The nodes are arranged in a complex, interconnected web-like pattern, with some nodes having multiple connections and others being isolated or part of small clusters.

Social Network:

An online platform or community where individuals or entities connect, communicate, and share information, often for social or professional purposes. Examples include Facebook, LinkedIn, and Twitter.

Transportation Network:

A system of roads, highways, railways, and other infrastructure that enables the movement of people and goods from one location to another.

Different Type of Networks

A decorative network diagram in the top right corner, featuring a complex web of interconnected nodes and lines, with some nodes highlighted in blue and others in grey.

Electrical Network:

A grid of interconnected electrical components (e.g., power generators, transformers, power lines) that distribute electricity to homes, businesses, and industries.

Communication Network:

A system that facilitates the transmission of information, such as telecommunication networks for phone calls, text messages, or data transmission.

Different Type of Networks

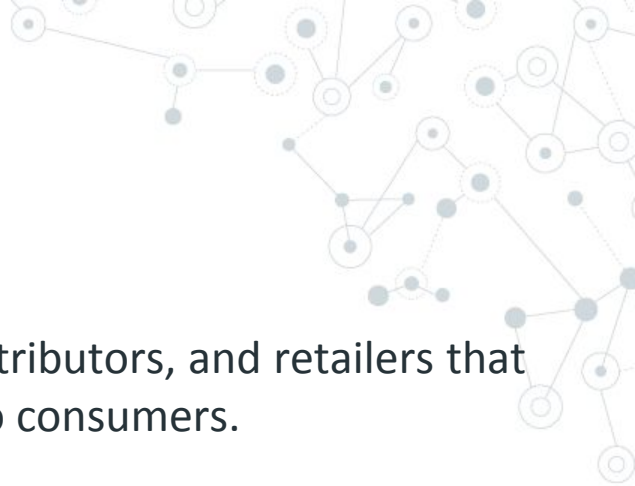
Neural Network:

A computational model inspired by the structure and function of the human brain, used in machine learning and artificial intelligence for tasks like pattern recognition and decision-making.

Biological Network:

A network of biological elements, such as neurons in the brain, genes in a genome, or species in an ecosystem, that interact to perform biological functions.

Different Type of Networks

A decorative network diagram in the top right corner, featuring a complex web of interconnected nodes and lines, with some nodes highlighted in blue and others in grey.

Supply Chain Network:

A network of organizations, suppliers, manufacturers, distributors, and retailers that collaborate to produce and deliver products or services to consumers.

Social Infrastructure Network:

A system of facilities and services (e.g., healthcare, education, utilities) that support and enhance the well-being of a community or society.

Different Type of Networks

Peer-to-Peer (P2P) Network:

A decentralized network architecture in which individual devices or nodes can directly communicate and share resources without relying on a central server.

Internet Applications

Communication:

- ☐ Line
- ☐ Email
- ☐ Skype
- ☐ Google Meet
- ☐ Zoom
- ☐ WhatsApp
- ☐ Facebook
- ☐ Messenger

Job Search:

- ☐ LinkedIn
- ☐ Glassdoor
- ☐ CareerBuilder
- ☐ ZipRecruiter

Online Shopping:

- ☐ Amazon
- ☐ eBay
- ☐ Paytm Mall
- ☐ Walmart
- ☐ AliExpress

Web Browsing:

- ☐ Internet Explorer
- ☐ Google Chrome
- ☐ Mozilla Firefox
- ☐ Safari
- ☐ Opera
- ☐ Conqueror
- ☐ Lynx

Internet Applications

Social Market Updates:

- ☐ Bloomberg
- ☐ CNBC
- ☐ Market Watch
- ☐ Investing.com
- ☐ Stock Twits
- ☐ Stock Edge

Travel:

- ☐ MakeMyTrip
- ☐ Expedia
- ☐ Airbnb
- ☐ Skyscanner
- ☐ Google Maps

Research:

- ☐ Google Scholar
- ☐ Mendeley
- ☐ Flipboard
- ☐ Researcher
- ☐ Google Keep
- ☐ Curiosity
- ☐ Paperity

E-Commerce:

- ☐ Amazon
- ☐ Paytm Mall

Online Payments:

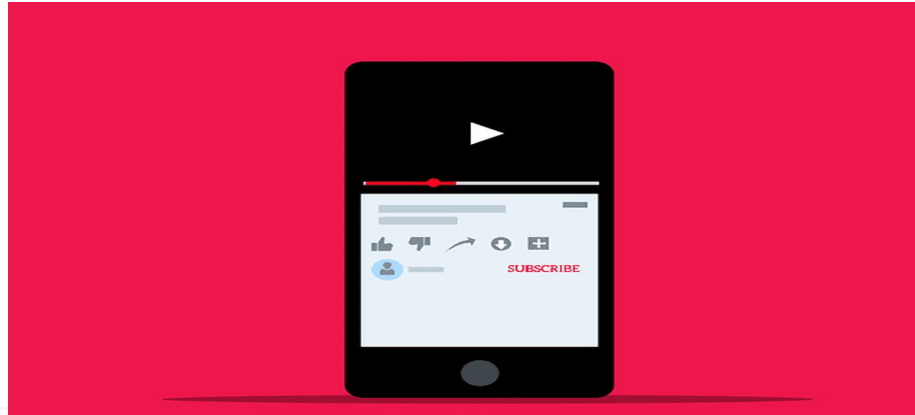
- ☐ Paypal
- ☐ Paytm
- ☐ Google Pay
- ☐ Amazon Pay

Internet Applications

Social Networking:



Entertainment:



Internet Applications

E-Banking:



Internet Applications

E-Learning:



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- ◎ Security
- ◎ Protocol layers, service models

Internet

Two ways to explain internet:

1. we can describe the nuts and bolts of the Internet, that is, the basic hardware and software components that make up the Internet.
2. Second, we can describe the Internet in terms of a networking infrastructure that provides services to distributed applications

The Internet: a “nuts and bolts” view



Billions of connected computing **devices**:

- **hosts** = end systems
- running **network apps** at Internet's “edge”

Packet switches: forward packets (chunks of data)

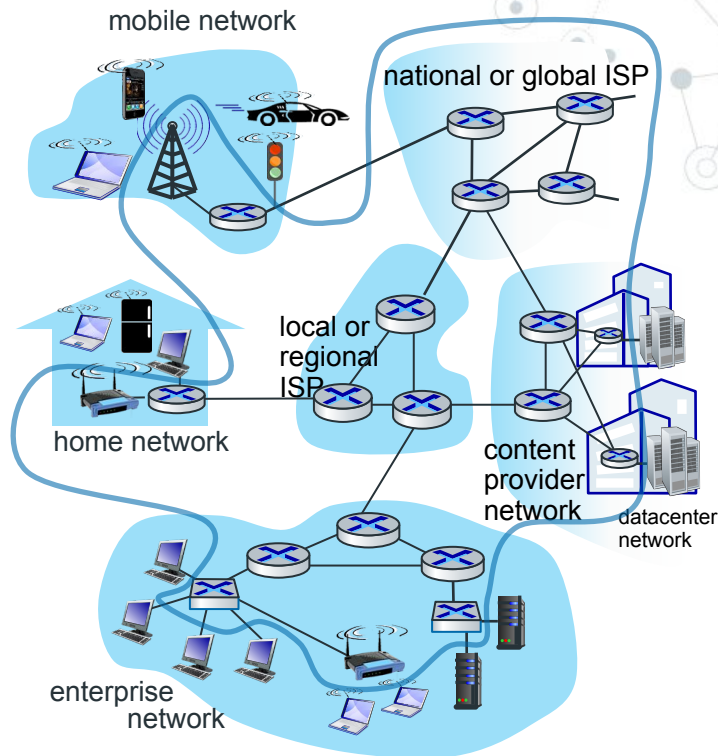
- **routers, switches**

Communication links

- fiber, copper, radio, satellite
- transmission rate: **bandwidth**

Networks

- collection of devices, routers, links: managed by an organization



“Fun” Internet-connected devices



Amazon Echo



Internet refrigerator



IP picture frame



Pacemaker & Monitor



Tweet-a-watt:
monitor energy use



Security Camera



Slingbox: remote
control cable TV



Web-enabled toaster +
weather forecaster



AR devices

Internet phones



sensorized,
bed
mattress



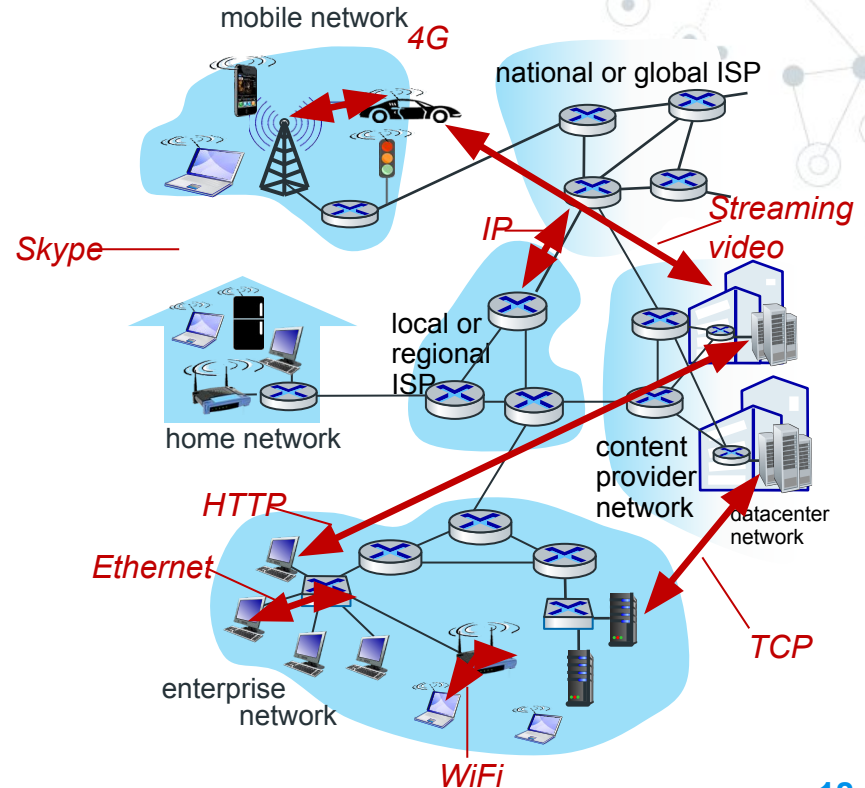
Fitbit

Others?

The Internet: a “nuts and bolts” view

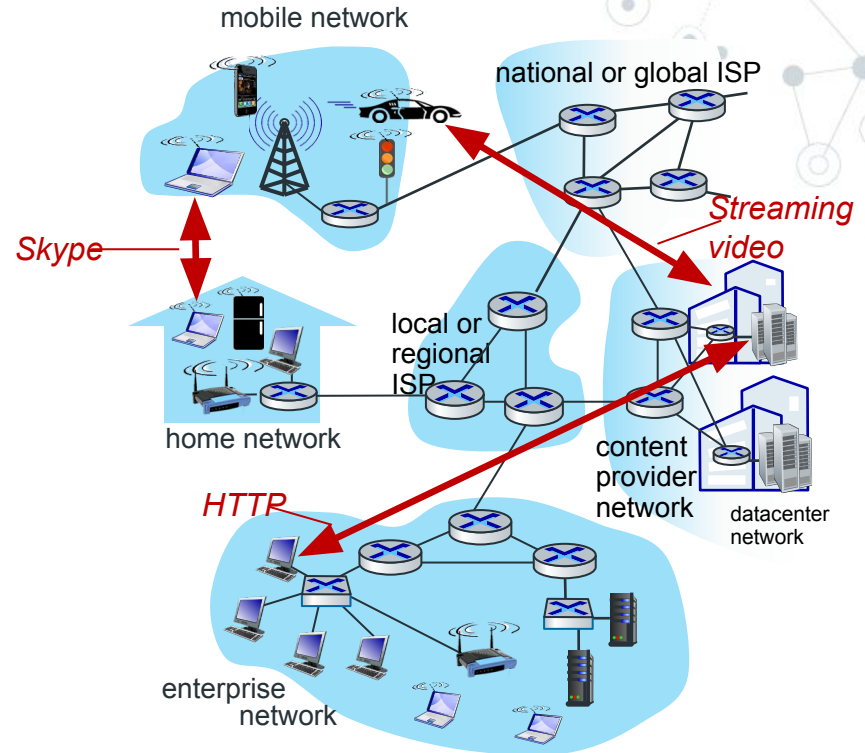
◎ *Internet: “network of networks”*

- Interconnected ISPs
- *protocols* are everywhere
 - control sending, receiving of messages
 - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet
- *Internet standards*
 - RFC: Request for Comments
 - IETF: Internet Engineering Task Force



The Internet: a “service” view

- **Infrastructure** that provides services to applications:
 - Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, ...
- provides **programming interface** to distributed applications:
 - “hooks” allowing sending/receiving apps to “connect” to, use Internet transport service
 - provides service options, analogous to postal service



The Internet: a “service” view

- **Analogous to postal service**
- Suppose Alice wants to send a letter to Bob using the postal service. Alice, of course, can't just write the letter (the data) and drop the letter out her window. Instead, the postal service requires that Alice put the letter in an envelope; write Bob's full name, address, and zip code in the center of the envelope; seal the envelope; put a stamp in the upper-right-hand corner of the envelope; and finally, drop the envelope into an official postal service mailbox. Thus, the postal service has its own “postal service interface,” or set of rules, that Alice must follow to have the postal service deliver her letter to Bob. In a similar manner, the Internet has a socket interface that the program sending data must follow to have the Internet deliver the data to the program that will receive the data.
- The postal service, of course, provides more than one service to its customers. It provides express delivery, reception confirmation, ordinary use, and many more services. In a similar manner, the Internet provides multiple services to its applications. When you develop an Internet application, you too must choose one of the Internet's services for your application.

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What's a protocol?

Human protocols:

- “what’s the time?”
- “I have a question”
- introductions

... specific messages sent

... specific actions taken when message received, or other events

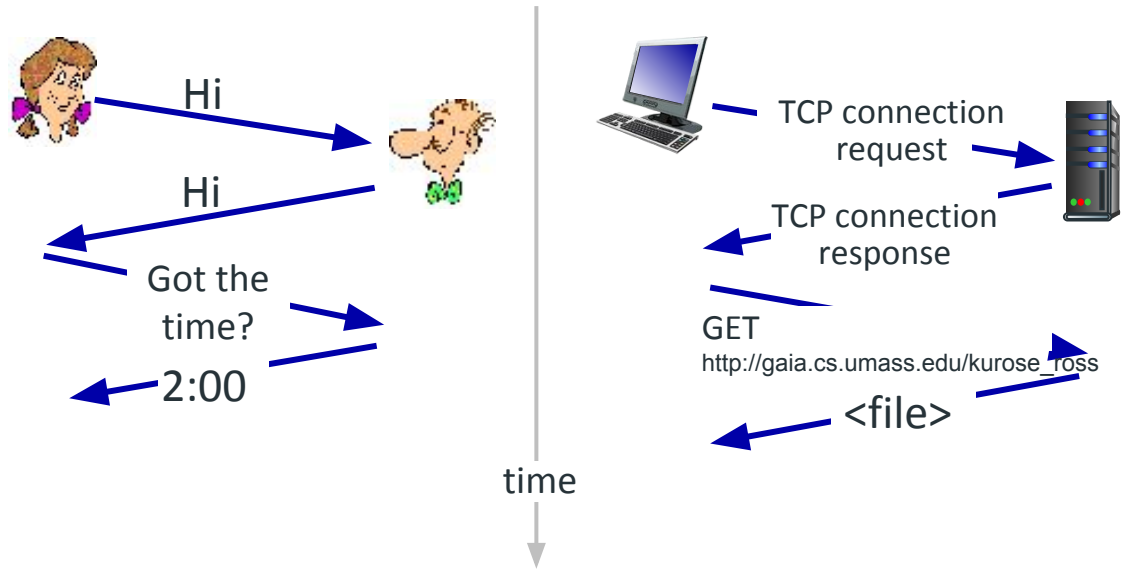
Network protocols:

- computers (devices) rather than humans
- all communication activity in Internet governed by protocols

Protocols define the format, order of messages sent and received among network entities, and actions taken on msg transmission, receipt

What's a protocol?

A human protocol and a computer network protocol:



Q: other human protocols?